If you encounter duplicate indexes, one of them can be removed.

Watch out for Index Duplicates

PostgreSQL Does Not Prevent Duplicate Indexes



- Duplicate Indexes can be created to cover the same fields as long as the indexes have different names.
- Remove duplicate Indexes

Exact duplicates are straightforward and not common. What is less straightforward and more common is overlapping or redundant Indexes.

In this scenario two or more indexes cover the same column and only one of them is necessary. The redundant index may have a different definition, but never be used by a query.

For example, when a query filters on the first_name column of the users table, a single column index covering first_name could be used. A Multicolumn index added later covering first_name and last_name could be used for filtering on first_name as well. These two indexes overlap, both covering the first_name field. If the Multicolumn index was added for a query later, it's possible the single column index is no longer needed.

You'll want to see if there are Index Scans for each of these Indexes. To help find these, you can use the PgHero tool. PgHero detects and displays overlapping indexes. The source code has a method duplicate_indexes that will compare columns being covered by indexes for a table and check for overlapping definitions.

This information is presented in a section on the main screen called *Duplicate Indexes*. For example, PgHero will explain the following.

> These indexes exist, but aren't needed. Remove them with a migration for faster writes.
>
> users_first_name_idx (first_name) is covered by users_first_name_email_idx (first_name, email)

In the example above, the second Multicolumn index can be used by queries for the first_name column. PgHero is suggesting that the first single column index on first_name is unnecessary. An Active Record Migration is even generated to perform the removal.

Bloat accumulates in tables with high amounts of Updates and Deletes. What about tables that only receive Inserts?